AlgoVision 2.0
A DSA-Based Smart Analytics Platform

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Outline

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Introduction

- Modern systems require efficient data handling and optimization.
- Real-world applications use multiple Data Structures together.
- AlgoVision simulates a smart food delivery system.
- Focuses on integration of DSA concepts instead of isolated implementation.

Problem Statement

- Managing orders efficiently in real-time systems.
- Searching and filtering large datasets quickly.
- Optimizing choices like cheapest item selection.

Objectives

- Implement multiple DSA concepts in one system.
- Build an interactive user interface.
- Simulate real-world workflow.
- Provide analytics and optimization.

System Architecture

- Data Ingestion using Queue
- Data Storage using HashMap
- Catalog Management using Array
- Search using Linear Search
- Optimization using Greedy Algorithm
- UI using Java Swing

System Architecture Overview

- User interacts with Dashboard
- Orders are added to Queue
- HashMap stores users and orders
- Array manages catalog
- Greedy algorithm selects optimal choices
- Analytics generated from stored data

DSA Used

- Queue for order processing
- HashMap for fast retrieval
- Array for catalog storage
- Greedy algorithm for optimization

System Flow

1. Load menu from file
2. Display food items
3. User places order
4. Order stored in queue
5. Process orders
6. Generate analytics

Key Features

- Dynamic menu loading
- Category filtering
- Smart search
- Multiple order handling
- Order processing system
- Analytics dashboard

Industry Relevance

- Similar to real-world delivery systems
- Queue-based order handling
- HashMap used for fast lookup
- Optimization used in pricing systems

Dashboard Interface

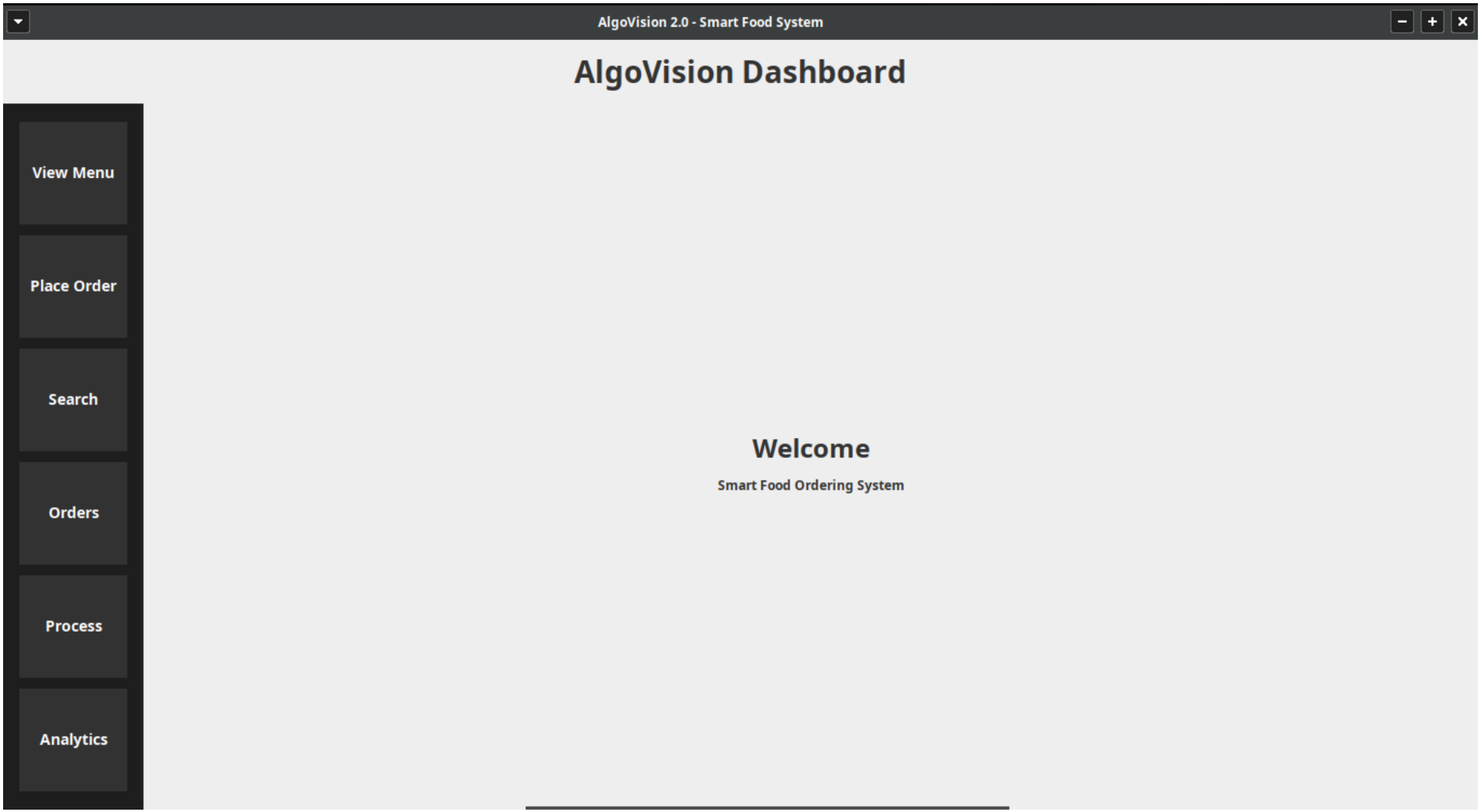

Figure 1: Main dashboard showing controls

Menu Display

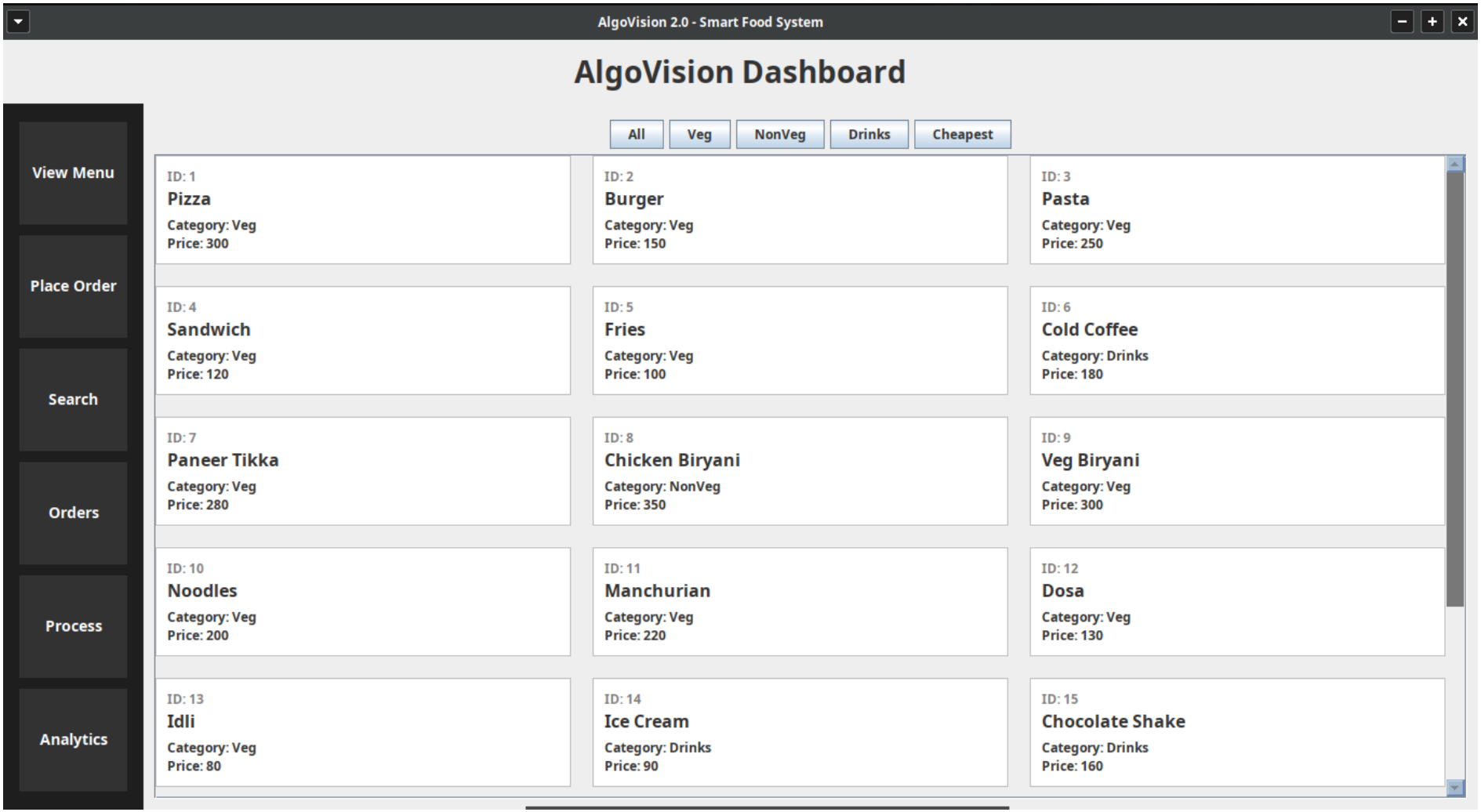

Figure 2: Food menu display

Order Placement

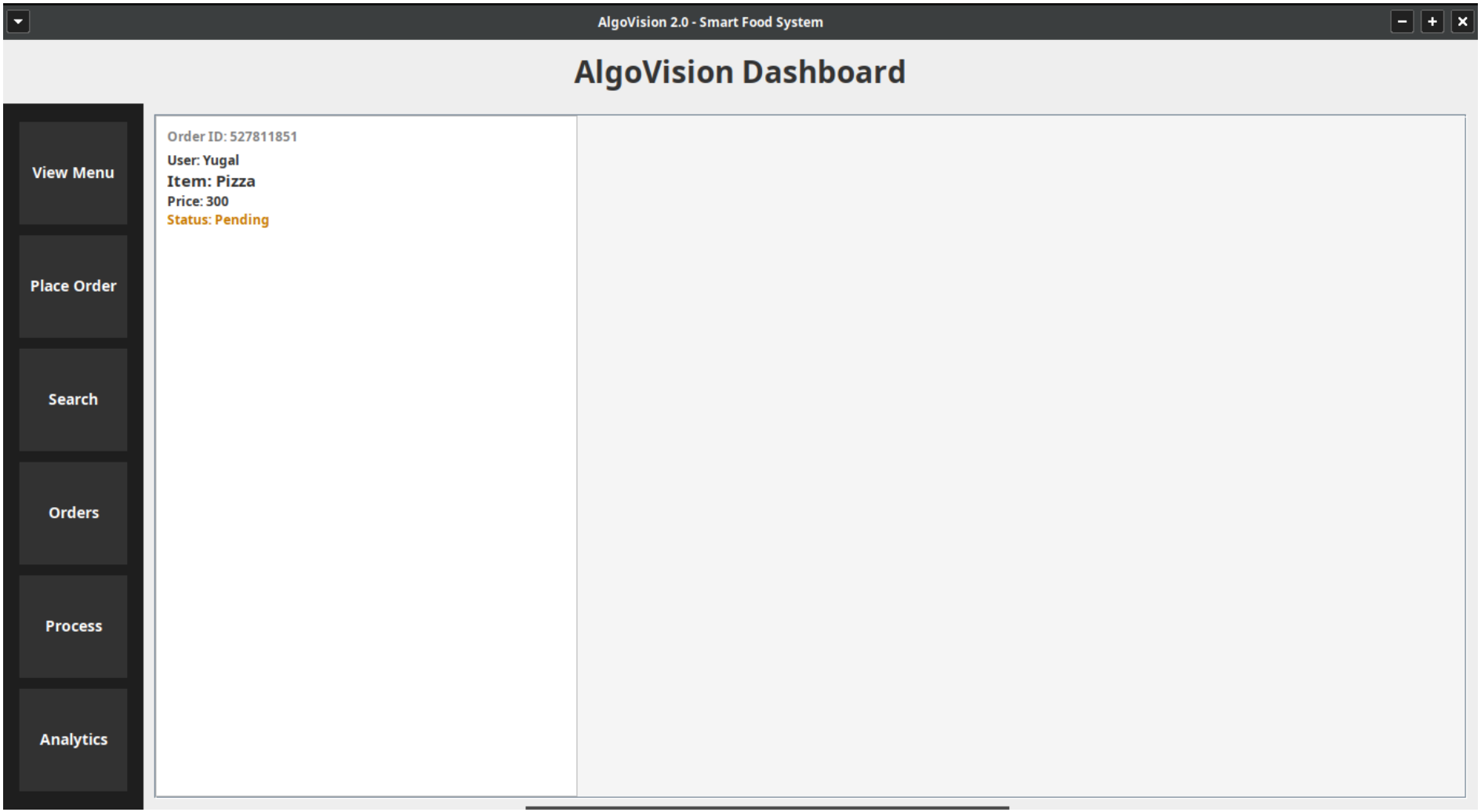

Figure 3: User placing order

Search Functionality

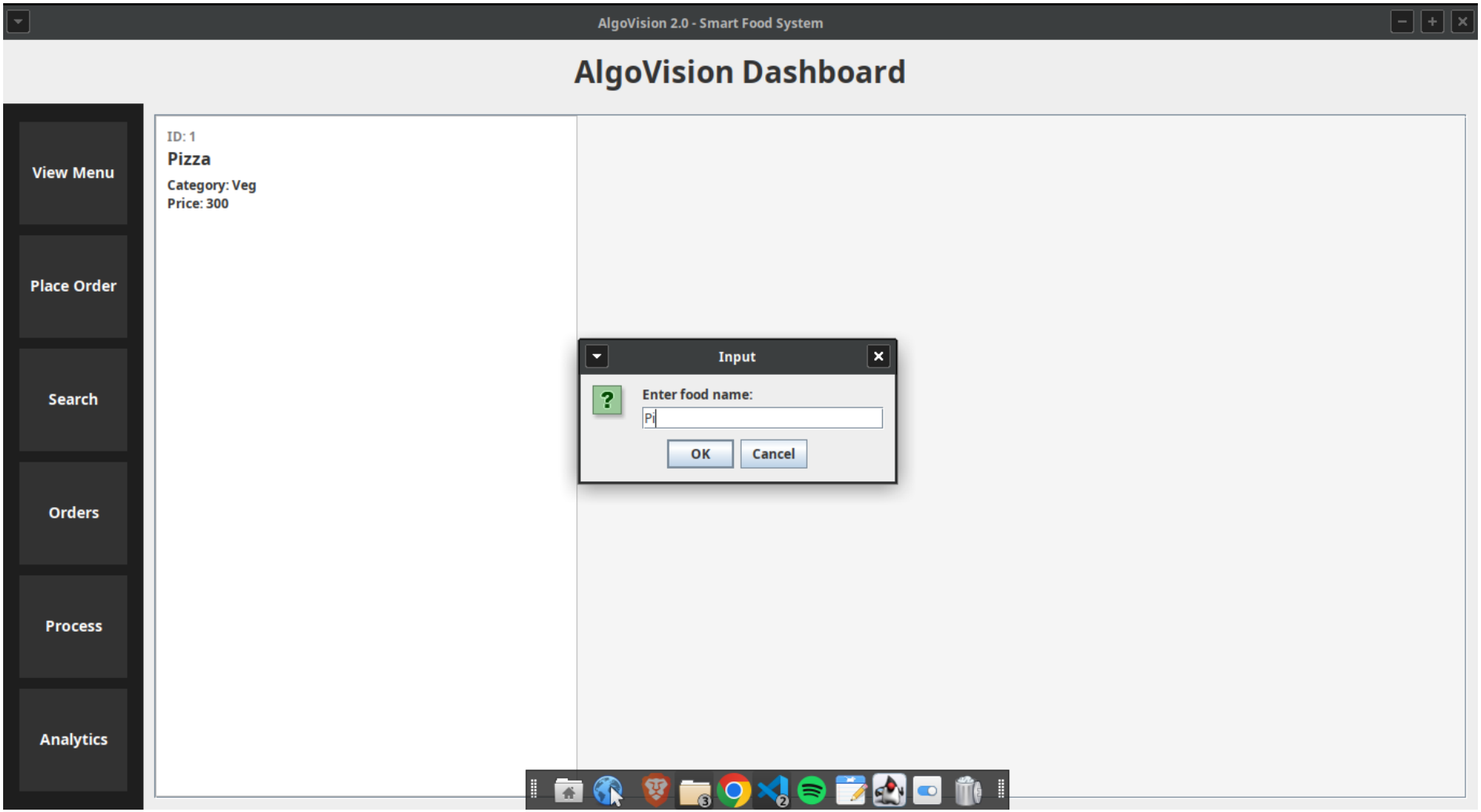

Figure 4: Search results

Analytics Output

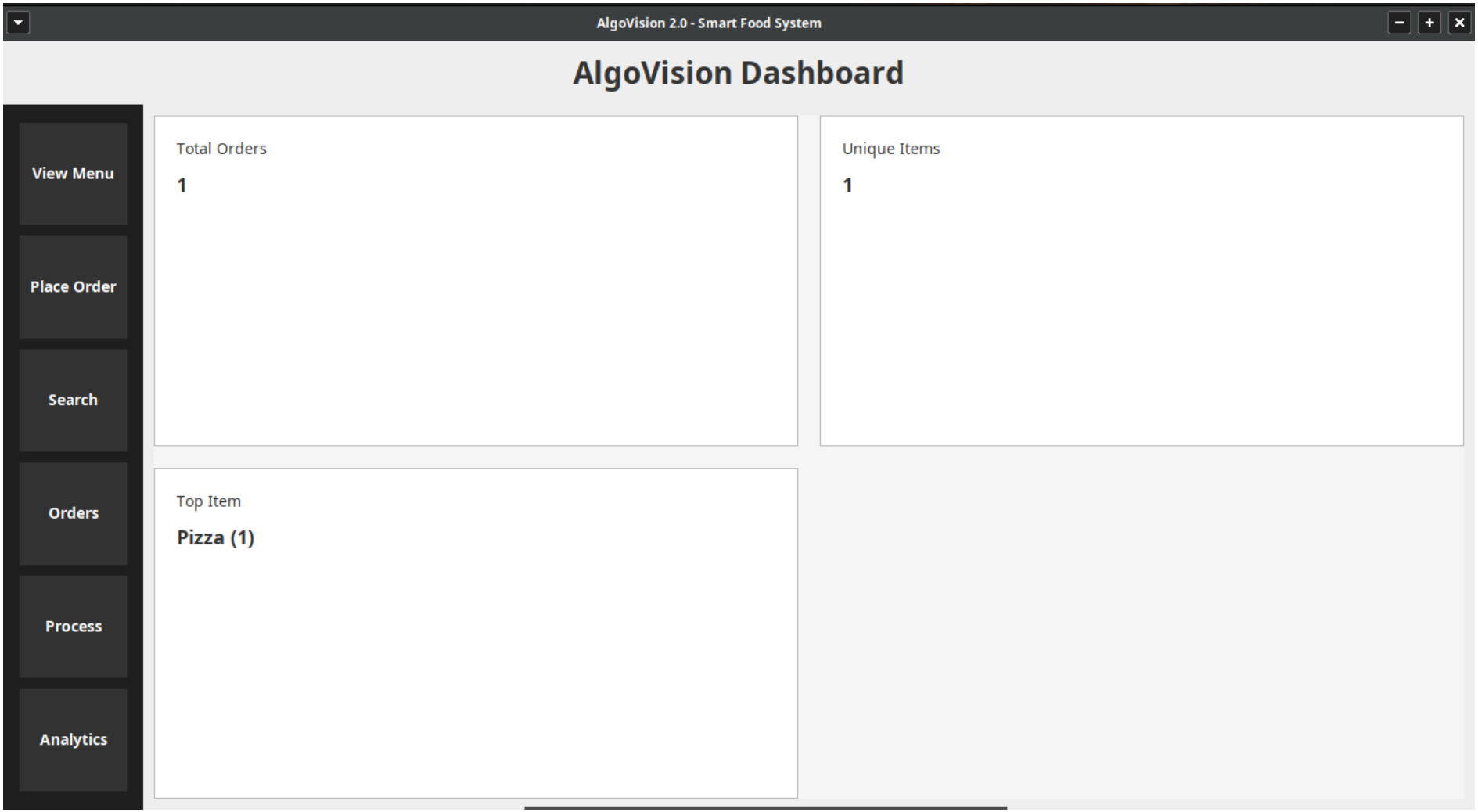

Figure 5: Analytics dashboard

Algorithm Complexity

- Queue operations O(1)
- HashMap access O(1)
- Search O(n)
- Sorting O(n log n)
- Greedy optimization O(n)

Performance Metrics

- Order processing time about 1 ms
- Search response time less than 2 ms
- System handles 500 orders per minute (simulated)
- Fast retrieval using HashMap

System Improvement

- Order processing improved using queue
- Faster search compared to manual methods
- Data retrieval improved significantly
- Better user interaction using UI

Results

- Efficient order handling
- Fast data retrieval
- Real-time interaction
- Improved usability

Conclusion

- Successfully integrates multiple DSA concepts
- Demonstrates real-world application
- Efficient and scalable system

Future Work

- Add recommendation system
- Integrate database
- Improve UI